

QueueLess

Smart Queue & Intelligent Waiting Management System



**A queue should reserve your turn,
not imprison your time.**

Track: Open Innovation / Everyday Automation

A human-centred problem statement for a real-world service challenge

1. The Problem: Waiting Has Become an Unnecessary Burden

Imagine an elderly person standing in a crowded government hospital, waiting for a consultation. They cannot leave the queue because they might miss their turn. They cannot comfortably stand for long periods, and they have no reliable information about how much longer they must wait.

Now imagine a student waiting outside an examination or administrative office, a customer waiting at a bank, or a passenger standing in a crowded service center.

Different places. Different people. But the same problem:

People are forced to organize their lives around a queue, instead of the queue adapting to their lives.

Traditional queue systems depend on physical presence, paper tokens, or manually managed lines. They provide little visibility into actual waiting times, unexpected delays, or changes in service speed.

As a result, people experience:

- Lost productive hours due to uncertain waiting.
- Physical discomfort from standing in heat, rain, dust, or crowded spaces.
- Anxiety about missing their turn if they step away for water, food, medication, or restroom use.
- Repeated visits because they cannot predict when service will be available.
- Congestion and frustration caused by everyone gathering at the service location simultaneously.

The problem is not simply that queues are long.

The real problem is that waiting is unpredictable, restrictive, and poorly managed.

2. Real-World Scenarios That Expose the Pain



Scenario 1: The Hospital Patient

An elderly patient arrives at a government hospital at 9 AM. There are 40 people ahead. They have no idea whether their consultation will happen in 30 minutes or three hours. They cannot leave to eat, rest, or collect medicine without risking their place.

Pain: Physical exhaustion, uncertainty, and unnecessary exposure to crowded waiting areas.

Scenario 2: The Bank Customer

A working professional visits a bank for a task that takes five minutes at the counter but spends nearly an hour waiting. They cannot leave because their number might be called at any moment.

Pain: A small service consumes a disproportionately large part of someone's working day.

Scenario 3: The Government Office Visitor

A citizen visits an administrative office for document verification. They receive a token but have no live information about delays or their expected service time. They remain inside a crowded building for hours, even when their turn is far away.

Pain: Lack of transparency turns a routine public service into an unpredictable day-long commitment.

Scenario 4: The Everyday Citizen

Someone waiting at a railway ticket counter, diagnostic lab, college office, or service center cannot confidently plan anything around their waiting time. Even a short break becomes a risk.

Pain: The queue takes control of the person's time, rather than simply organizing service delivery.

3. The Core Problem We Are Solving

Current queue systems primarily answer:

“Who is next?”

But they fail to answer the questions people actually need:

- How long will I realistically have to wait?
- Can I leave the premises temporarily without losing my place?
- When should I return?
- What happens if service is delayed?
- Can the system adapt when the queue suddenly becomes longer or shorter?

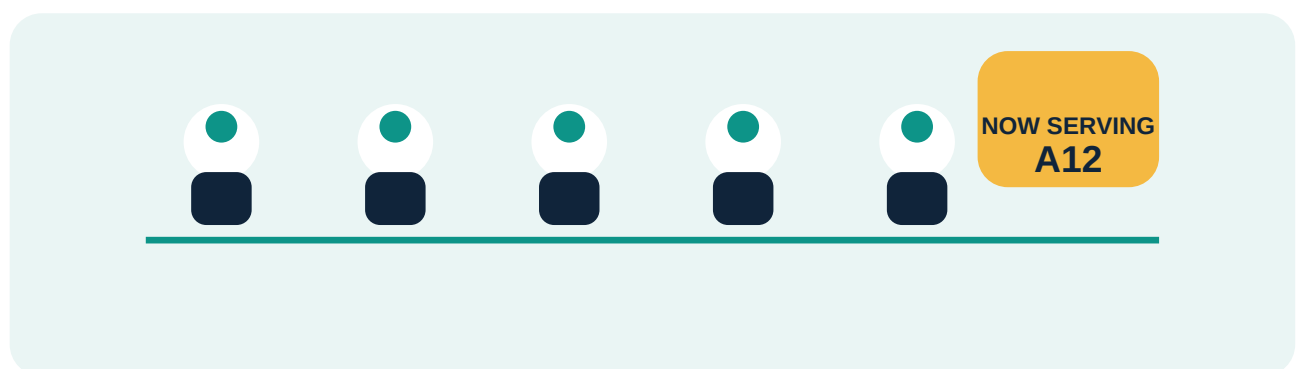
This creates a fundamental gap between queue management and actual human needs.

4. Our Proposed Solution

We propose an intelligent, real-time queue management system that transforms passive waiting into predictable, flexible waiting.

Instead of forcing people to remain physically present, the system provides:

- Digital queue tokens.
- Live queue position and estimated waiting time.
- Dynamic wait-time recalculation as service progresses.
- Notifications when a person's turn approaches.
- The ability to wait elsewhere and return at the appropriate time.
- A service-provider dashboard to monitor queue flow and identify bottlenecks.



The system aims to make waiting more transparent, accessible, and manageable across hospitals, banks, diagnostic centers, government offices, educational institutions, and other high-footfall service locations.